

4. Project Planning

FitBuddy — Development phases, milestones, deliverables, and verification plan

Implementation Phases

Phase 1 — Foundation

Create the FastAPI application, dependency configuration, project structure, database connection, and base models.

Phase 2 — Requirements & Validation

Implement Pydantic schemas, accepted goal/intensity/experience values, field limits, and form validation behavior.

Phase 3 — Core Generation

Connect the workout and nutrition generation wrappers, store generated results, and render the result page.

Phase 4 — Feedback Loop

Retrieve the latest plan, submit user feedback to the update service, persist the revised plan, and show the updated result.

Phase 5 — Administration & APIs

Implement user listing, deletion, health monitoring, and JSON endpoints.

Phase 6 — Testing & Documentation

Run endpoint tests, verify fallback behavior, document setup, and prepare demonstration material.

Milestone Checklist

- M1: Project boots and home page renders.
- M2: User profile validates and persists.
- M3: Workout and nutrition outputs are generated and saved.
- M4: Feedback updates the latest plan.
- M5: Admin dashboard and delete operation work.
- M6: API tests pass and README setup instructions are complete.
- M7: Final demonstration can run with or without a configured Gemini key.

Risk and Mitigation

- AI service unavailable → use the implemented local fallback for development/testing.
- Invalid user input → enforce Pydantic constraints and display form errors.

- Database state affects repeated tests → use predictable test identifiers and reset/isolated test database strategy for larger deployments.
- Model behavior changes → keep model names configurable through environment settings.
- Wellness misuse → present the product as wellness guidance, not medical diagnosis or treatment.

Suggested Project Schedule

For an academic project, the phases can be mapped to eight working sessions: ideation, requirements, design, planning, development, testing, documentation, and final demonstration. Each session should end with a concrete artifact that can be reviewed before moving forward.

Quick Reference

Milestone	Primary Deliverable	Verification
M1	Running skeleton	GET / returns 200
M2	Validated profile	Invalid values rejected
M3	Generated plan	Workout + tip stored
M4	Feedback revision	Updated plan returned
M5	Admin/API layer	Users visible and deletable
M6	Test suite	pytest passes
M7	Demo package	README + presentation flow

Project basis: This document is derived from the supplied FitBuddy source package, including its README, FastAPI routes, SQLAlchemy models, Jinja2 templates, CSS, and tests.